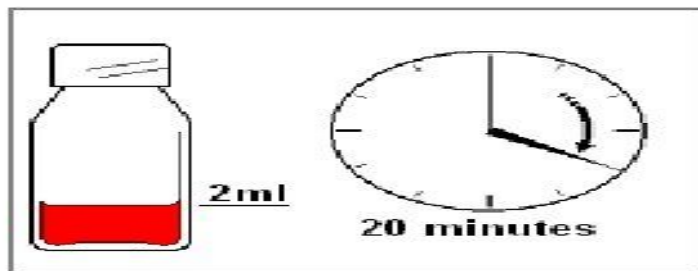


# CLOTTING TIME

## Clotting Time using Test Tube Method



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## **OBJECT:**

**To determine your own clotting time**

# OBJECTIVES:

- The student should be able to:
  - Find out the clotting time
  - Know the causes which may lead to prolonged and defective clotting time
  - Have knowledge about anticoagulants used in the laboratory
- 
- Students should know about the ,
  - Hemophilia
  - Thrombosis
  - Thromboembolic condition and their hazards
  - Differentiate between bleeding time and clotting time

# REQUIREMENTS:

- Capillary tube
- Stop watch and
- Pricking needle

## PRINCIPLE:

- Principle of the experiment is to observe the time when the insoluble monomer fibrin clot is formed

# THEORETICAL EXPLANATION:

- Whenever there is cut or injury the blood comes in contact with rough surface and the pro-coagulants present in blood are activated and convert the plasma protein **Prothrombin** into **Thrombin** which converts the soluble plasma protein **Fibrinogen** into an insoluble protein **Fibrin**, jelly like semisolid substance called **clot**.
- This is a **loose** type of plug and may be dislodged by the pressure of the blood stream. The blood clot then contracts and firmly plugs the rent in the vessel
- This retraction of blood clot is the next step of hemostasis

# THEORETICAL EXPLANATION:

- Clotting time is the time from the time of injury to the formation of clots i.e. **Fibrin Thread**
- Normal clotting time is **2-8 min**

# CLINICAL SIGNIFICANCE:

- **Excessive bleeding tendencies** are related to the **clotting factor defect** and the deficiency and the **defect of platelets**.
- Clotting time is **increased** due to **decrease in clotting factor**, **factor viii deficiency** causes **hemophilia** etc.
- The hemorrhagic diseases due to selective deficiencies of other clotting factors are:
- **Afibrinogenemia** is the **lack of factor 1** (fibrinogen). It may be due to **complicated pregnancies**
- Fibrinogen is formed in the **liver** therefore it is reduced in **liver disorders**



# CLINICAL SIGNIFICANCE:

- **Hypo-prothrombinemia** (bleeding tendency in liver disease caused by lack of factor 2, (Factor-II, prothrombin),: occurs due to **decreased hepatic synthesis of prothrombin** usually secondary to **vitamin K deficiency**
- **Para hemophilia** is caused by deficiency of **factor V** (labile factor) which is congenital
- Disorder caused by deficiency of factor VII (Proconvertin)
- **Hemophilia A (classical)** caused by **factor VIII deficiency (antihaemophilic factor)** is relatively common which may be treated by repeated transfusion of blood or factor VIII rich preparation of plasma. It is an inherited sex link recessive disorder

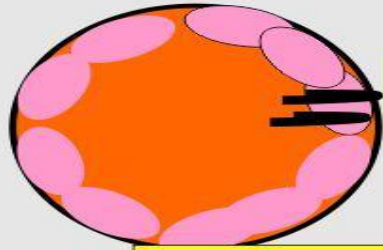
## CLINICAL SIGNIFICANCE:

- **Hemophilia B (Christmas)** is also a congenital disease caused by deficiency of **factor IX**
- The **congenital bleeding tendencies** are also caused due to deficiency **factors X, XI and XII**
- In **obstructive jaundice**, absorption of **vitamin K** along with other fat soluble vitamins is suppressed. The resulting clotting factor deficiencies cause **bleeding tendency**

# CLINICAL SIGNIFICANCE:

- The **Thromboembolic condition** is an **intravascular clotting**
- **Thrombosis** may occur due to **deposition of cholesterol** (arteriosclerosis) i.e. formation of plaques
- The Thrombus may get detached and lead to **embolism** which may be **pulmonary embolism**, **coronary embolism** which may lead to **ischemia** and **heart attack**
- The **emboli** may **block the arteries** in **brain**, **kidneys** or elsewhere

## Bleeding control mechanism



Injury to vessel

Contraction of the vessel

Adherence of the platelets

Activation of the clotting factor

Formation of the clot

Bleeding stop

Prolonged bleeding time if

- ① Low platelets count
- ② Inadequate platelets function
- ③ Inadequate vessel contraction

# PROCEDURE:

- Prick the ring finger with disposable blood lancet after sterilization of the finger by spirit swab and start the stopwatch. Discard the first drop of blood
- Gently press the finger and allow the blood to come out freely & touch one end of capillary tube with the blood drop by holding the capillary tube horizontally and allow the tube to fill by capillarity. Capillary should be placed in the middle of blood drop
- Fill the capillary tube with blood which appears in the tube by capillary action
- Start breaking a small piece of tube at distance of 1 cm, separate the broken piece very gently and observe for the formation of fibrin thread

## PROCEDURE:

- Repeat the process after every 30 sec observing each time for the appearance of fibrin thread
- End point is reached when fibrin thread appears
- Stop the stop watch
- Note the clotting time on the observation chart and draw the figure of tube

# PRECAUTIONS:

- **No air bubble** should **interrupt** the column of blood in capillary tube
- **The blood** should be **immediately filled in the tube** to avoid clotting of blood before filling of tube
- The **stop watch** should **be started immediately** as the prick is made
- A **small piece of tube** (about 1cm) should be **first broken** & then be **separated very carefully** to avoid breaking of fibrin thread

# PRECAUTIONS:

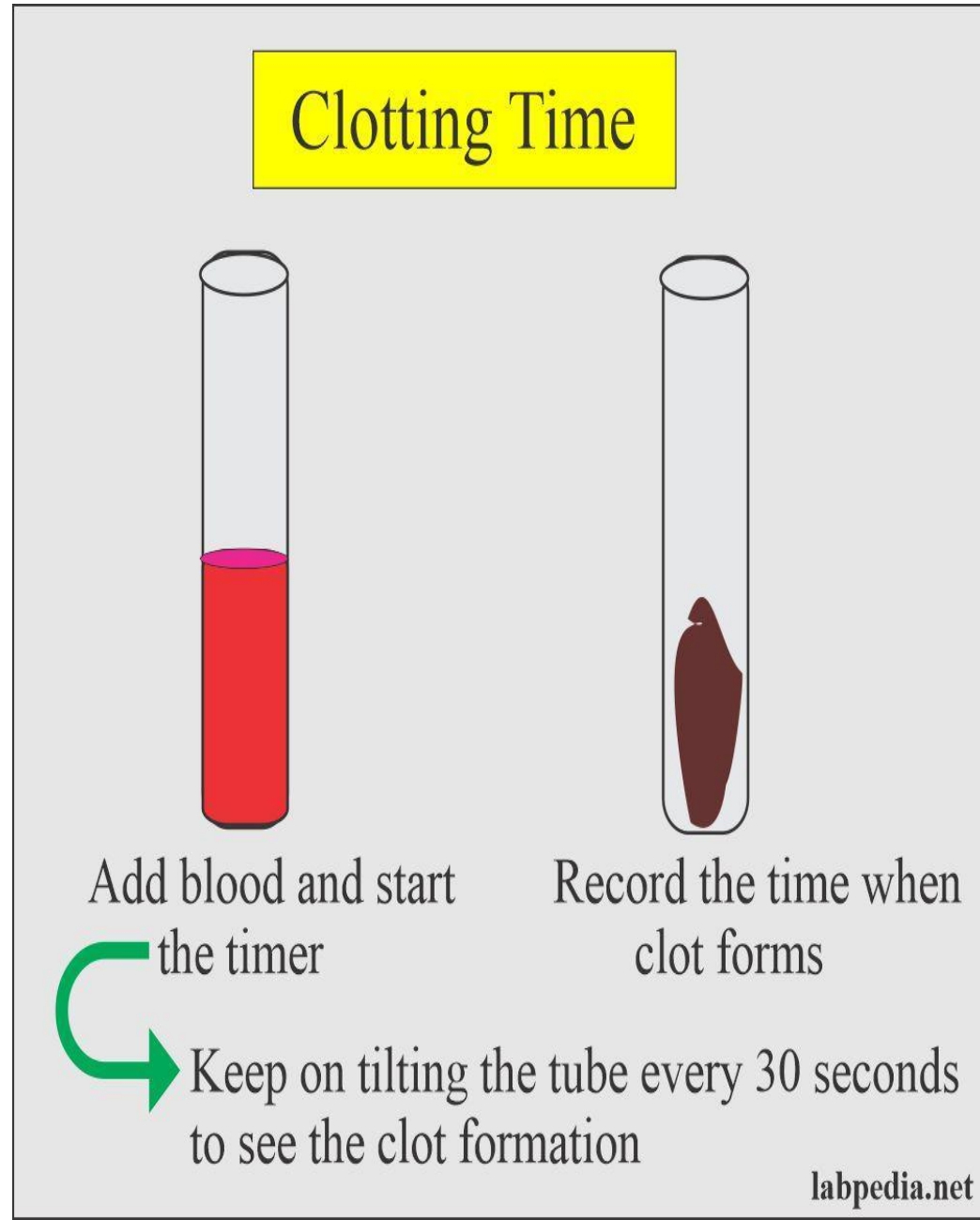
- **IMPORTANT:**
- **Don't use the 1<sup>st</sup> drop** of blood because it is clotted unusually fast
- **Note the time** that the **drop of blood to be used appears** at the site of puncture. This is used as the starting time
- **The tube** should be **kept at body temperature** as cold retards coagulation of blood (if the outside temperature is different then the tube can be placed in a water container at 37 °C)



# FIGURE

## OBSERVATION:

- **RESULT:**
- My clotting time is \_\_\_\_\_ sec



## **Q-1-What is clotting time**

This is an in vitro test and is the time taken for a blood sample to clot in a glass tube at 37°C. Normal values are between 4.5 and 7 minutes.

## **Q-2-Is there any other method (Than Capillary tube for determination of clotting time**

There are several other methods to determine clotting time besides the capillary tube method.

These include the Lee and White method,  
activated partial thromboplastin time (aPTT), prothrombin time (PT),  
activated clotting time (ACT),  
and thrombin time (TT)

## **Q-3-What are clotting factors**

Clotting factors are proteins in blood plasma that prevent excessive bleeding. They're essential to the cascade of events called coagulation

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Clotting factors are proteins in blood plasma that prevent excessive bleeding. They're essential to the cascade of events called coagulation

- The clotting factors are
- Factor I (fibrinogen),
- Factor II (prothrombin),
- Factor III (tissue thromboplastin or tissue factor),
- Factor IV (ionized calcium),
- Factor V (labile factor or proaccelerin),
- Factor VII (stable factor or proconvertin),
- Factor VIII (antihemophilic factor).
- Factor IX (plasma thromboplastin component or the Christmas factor),
- Factor X (Stuart-Prower factor),
- Factor XI (plasma thromboplastin antecedent),
- Factor XII (Hageman factor),
- Factor XIII (fibrin-stabilizing factor).

#### **Q-4-Why blood does not clot within the blood vessels normally?**

Blood generally does not clot within blood vessels due to a combination of factors that maintain a state of anticoagulant balance. These include the presence of anticoagulants like heparin, the smooth, non-thrombogenic surface of the endothelium, and the continuous, rapid flow of blood.

#### **Q-5-What initiates the process of clotting?**

The process of blood clotting is initiated by damage to a blood vessel, triggering a cascade of reactions involving platelets and clotting factors. When a blood vessel is injured, platelets adhere to the site of damage and release chemicals to attract more platelets, forming a plug.

#### **Q-6-What are the naturally occurring anticoagulant in blood?**

Naturally occurring anticoagulants in blood include proteins like antithrombin III, protein C, protein S, and tissue factor pathway inhibitor (TFPI). These substances help regulate the blood clotting process and prevent excessive or inappropriate clot formation.

### **Q-7-Which anticoagulant is used in the laboratory?**

In the laboratory, common anticoagulants include EDTA, heparin, and citrate. These are used to prevent blood samples from clotting, allowing for various tests and analyses.

### **Q-8-What is the role of calcium in the mechanism of coagulation?**

Calcium ions ( $\text{Ca}^{2+}$ ) play a crucial role in the coagulation cascade, acting as a cofactor in several key steps. They are essential for blood clotting by facilitating the activation of various coagulation factors and mediating the conversion of fibrinogen to fibrin, which forms the structural basis of a blood clot.

### **Q-9-What is the significance of determining clotting time?**

Determining clotting time is crucial for assessing a patient's ability to stop bleeding after an injury and for diagnosing potential bleeding or clotting disorders. It helps doctors understand how efficiently the body's natural clotting process works, which is essential for preventing excessive blood loss or, conversely, the formation of harmful clots.

## **Q-10-What are the clinical conditions in which clotting time is prolonged?**

Prolonged clotting time, indicating a longer than normal time for blood to clot, can occur in various clinical conditions.

These include inherited bleeding disorders like hemophilia and von Willebrand disease, acquired conditions like liver disease, vitamin K deficiency, and disseminated intravascular coagulation (DIC).

Certain medications, such as anticoagulants, can also prolong clotting time.